

Name: _____

Class: _____

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Year 10 2026

Binary Addition

Work in pencil. One column at a time, right to left — the 1s column first. A carry is written in the **next column to the left**, because that is the column it gets added into. In the four rules below, a **ringed** digit is a carry.

The four rules

CARRY

	0
	0
	0

$$0 + 0 = 0$$

CARRY

	0
	1
	1

$$0 + 1 = 1$$

CARRY

	1
	1
①	0

$$1 + 1 = 0 \text{ carry } 1$$

CARRY

	1
	1
	①
①	1

$$1 + 1 + 1 = 1 \text{ carry } 1$$

Worked example 43 + 74

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	1	0	1	0	1	1	43
Number	0	1	0	0	1	0	1	0	74
Answer	0	1	1	1	0	1	0	1	117
Carries				1		1			

Read it right to left. In the 2s column $1 + 1 = 0$ carry 1, so a **1** goes in the Carries row under the 4s column and is added in there. The denary column is the check: $43 + 74 = 117$, and the answer row reads 117.

Your turn

1. Write **12** and **7** into the grid in binary, then add them.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	0	0	1	1	0	0	12
Number	0	0	0	0	0	1	1	1	7
Answer	0	0	0	1	0	0	1	1	19
Carries				1	1				

2. Write **46** and **67** into the grid in binary, then add them.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	0	1	0	1	1	1	0	46
Number	0	1	0	0	0	0	1	1	67
Answer	0	1	1	1	0	0	0	1	113
Carries				1	1	1			

3. Write **89** and **52** into the grid in binary, then add them.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	1	0	1	1	0	0	1	89
Number	0	0	1	1	0	1	0	0	52
Answer	1	0	0	0	1	1	0	1	141
Carries	1	1	1						

4. The two binary numbers are already in the grid. Add them, then fill in the denary column as a check.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	1	0	1	1	0	1	0	90
Number	0	0	1	0	1	1	0	1	45
Answer	1	0	0	0	0	1	1	1	135
Carries	1	1	1	1					

5. Write **107** and **93** into the grid in binary, then add them.

Place value	128	64	32	16	8	4	2	1	Denary
Number	0	1	1	0	1	0	1	1	107
Number	0	1	0	1	1	1	0	1	93
Answer	1	1	0	0	1	0	0	0	200
Carries	1	1	1	1	1	1	1		

6. Write **150** and **130** into the grid in binary, then add them.

Place value	128	64	32	16	8	4	2	1	Denary
Number	1	0	0	1	0	1	1	0	150
Number	1	0	0	0	0	0	1	0	130
Answer	0	0	0	1	1	0	0	0	24
Carries					1	1			

6(b) Check your answer against the denary column. Something has gone wrong. What is this called, and why has it happened? **Overflow.** $150 + 130 = 280$, and 280 needs nine bits. Only eight columns exist, so the 1 that should sit in a 256s column has nowhere to go and is lost — the grid reads 24, which is $280 - 256$. The processor would set an overflow flag.